

Midterm 1 Practice Problems

Give complete justifications, including the relevant hypotheses when applying a theorem. You may quote results proved in class unless a proof is requested.

1. GCD and Euclidean algorithm

- (a) Use the Euclidean algorithm to compute $\gcd(323, 119)$ and express the gcd as an integer linear combination of 323 and 119.
- (b) Using your identity, find all integer solutions $(x, y) \in \mathbb{Z}^2$ of

$$323x + 119y = 85.$$

2. Unique factorization in $\mathbb{Z}$, I

Let k be a positive integer and let a, b be positive integers such that $\gcd(a, b) = 1$. Show that ab is a k th power of an integer if and only if a and b are both k th powers of integers.

3. Unique factorization in $\mathbb{Z}$, II

Let a, b be positive integers. Show that

$$\gcd(a, b) \cdot \text{lcm}[a, b] = ab,$$

where $\text{lcm}[a, b]$ is the least (positive) common multiple of a and b .

4. Linear congruences

- (a) Find all solutions of $42x \equiv 30 \pmod{96}$ in the range $0 \leq x < 96$.
- (b) Determine whether $42x \equiv 25 \pmod{96}$ is solvable, and justify your answer.

5. Fermat's little theorem

Let p be a prime satisfying $p \equiv 2 \pmod{3}$, let $a \in \mathbb{Z}$, and put

$$d = \frac{2p-1}{3}.$$

Show that d is a positive integer and that a^d is the unique solution of $x^3 \equiv a \pmod{p}$.

6. Euler's theorem at a prime power

Find the least nonnegative residue of

$$7^{110} + 10^{55} \pmod{81}.$$

7. Factorials and Wilson's theorem

Find the least nonnegative residue of

$$\frac{36!}{5 \cdot 8} \pmod{37}.$$

8. Legendre symbols and the quadratic reciprocity

Evaluate the following Legendre symbols and rephrase your answers in terms of quadratic residues and non-residues.

$$(a) \left(\frac{33}{107} \right), \quad (b) \left(\frac{-35}{61} \right).$$

9. Quadratic congruences

Let p be an odd prime with $p \neq 7$. Determine exactly which residue classes of p modulo 7 make

$$x^2 + x + 2 \equiv 0 \pmod{p}$$

solvable.

10. An application of multiplicative inverses

(a) You are given

$$17^2 \equiv 11 \cdot 2^2 \pmod{35}.$$

Use a multiplicative inverse of 2 modulo 35 to find one integer x satisfying $x^2 \equiv 11 \pmod{35}$. Verify your answer.

(b) A student observes that $4^2 \equiv 3 \cdot 4^2 \pmod{8}$ and concludes, by dividing by 4^2 , that 3 is a square modulo 8. Explain the error and show that the conclusion is false.

11. Sums of two squares, I

Let S denote the set of integers that can be expressed as sums of two squares. Let a, b be positive integers such that $ab \in S$. Show that $a, b \in S$ if and only if $\gcd(a, b) \in S$, or equivalently, if and only if $\text{lcm}[a, b] \in S$.

12. Sums of two squares, II

For each integer $k \geq 0$, put

$$N_k = 2^k 3^{k+1} \cdot 13.$$

- (a) Determine exactly which values of k make N_k a sum of two integer squares.
- (b) For the smallest such k , give an explicit representation of N_k as a sum of two integer squares.

13. The ring $\mathbb{Z}[mi]$

Show that given any integer $m > 1$, the element $1 + mi$ is a prime in $\mathbb{Z}[mi]$.

14. Non-unique factorization in $\mathbb{Z}[\sqrt{-11}]$

Consider $R = \mathbb{Z}[\sqrt{-11}]$.

- (a) Prove that 3 is a prime in R .
- (b) Verify that 3 divides $(1 + \sqrt{-11})(1 - \sqrt{-11})$ in R , but divides neither $1 + \sqrt{-11}$ nor $1 - \sqrt{-11}$.

15. The ring $\mathbb{Z}[\sqrt{-2}]$

Determine all integers m for which

$$(2 + \sqrt{-2})\alpha = m + \sqrt{-2}$$

has a solution $\alpha \in \mathbb{Z}[\sqrt{-2}]$.